

Recap Checkpoint 3 — after 1.4 (Data Types, Data Structures & Boolean Algebra) — cumulative 1.1–1.4 — ANSWER SHEET

Separate answer sheet / mark scheme — keep for marking.

Mark scheme

Total = 35 marks. Award marks for valid alternatives in line with OCR positive-marking. All calculations checked below.

Q1 — [2] (AO1) — 1.1.2 - CISC has a **large/complex instruction set** with variable-length, multi-cycle instructions (complexity in hardware) (1). - RISC has a **small/simple instruction set** of fixed-length, single-cycle instructions (complexity in software/compiler) (1).

Q2 — [3] — 1.2.1 - Virtual memory uses **secondary storage as if it were RAM** (1)... - ...so programs **larger than physical RAM, or more programs than fit in RAM, can run** (1). (AO1) - Drawback (1): **disk thrashing** — heavy paging in/out slows the system because secondary storage is much slower than RAM. (AO2)

Q3 — [3] (AO1) — 1.2.2 - Correct order (2; 1 mark if one pair transposed): **lexical analysis** → **syntax analysis** → **code generation** → **optimisation**. - The stage that removes redundant instructions / improves efficiency is **optimisation** (1).

Q4 — [3] (AO2) — 1.3.1 - Key difference (1): **hashing is one-way / irreversible; encryption is two-way / reversible** with a key. - Passwords are **hashed** because they **never need to be recovered** — login only compares hashes, so a stolen database does not reveal plaintext (1). - Transmitted data is **encrypted** because the **recipient must recover (decrypt) the original data**, which requires a reversible process (1).

Q5 — [3] — 1.4.1 (a) $89 = 64 + 16 + 8 + 1 \rightarrow$ **0101 1001** (1). (AO2) (b) Nibbles **0101** = 5, **1001** = 9 → **0x59** (1 for nibble split, 1 for answer = 2). (AO2) Check: $5 \times 16 + 9 = 80 + 9 = 89$ ✓.

Q6 — [4] (AO2) — 1.4.1 (a) $39 =$ **0010 0111**; $52 =$ **0011 0100** (1). (b) -52: invert **0011 0100** → **1100 1011**, add 1 → **1100 1100** (1). (c) Add **0010 0111 + 1100 1100**:

```
  0 0 1 0 0 1 1 1    (39)
+ 1 1 0 0 1 1 0 0    (-52)
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  1 1 1 1 0 0 1 1
```

Result **1111 0011** has MSB = 1 → negative; magnitude = invert + 1 = **0000 1100 + 1 = 0000 1101** = 13, so the answer is **-13** (1 for the addition, 1 for converting back to denary). Check: $39 - 52 = -13$ ✓.

Q7 — [3] — 1.4.1 (a) Mantissa must start **0.1...**; **0.0001101** so shift the point **right by 3 places** → **0.1101** (1). Moving the point right by 3 **reduces the exponent by 3**: $4 - 3 = 1$, giving **0.1101×2^1** (1). (AO2) (b) Normalising removes wasted leading zeros so the mantissa holds the **maximum significant**

figures / greatest precision for the available bits (1). (AO1) Examiner tip: moving the point right **DECREASES** the exponent; the value is unchanged.

Q8 — [3] (AO2) — 1.4.3 - Distribute (distributive law): $A \cdot (\bar{A} + B) = (A \cdot \bar{A}) + (A \cdot B)$ (1). - $A \cdot \bar{A} = 0$ (complement / annihilation law) (1). - $0 + (A \cdot B) = A \cdot B$ (identity law) (1). Final answer: **$A \cdot B$** .

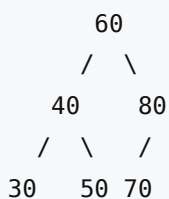
Q9 — [4] (AO2) — 1.4.3

A	B	F
0	0	0
0	1	1
1	0	1
1	1	0

(Working: $A+B$ is 0,1,1,1; $\neg(A \cdot B)$ is 1,1,1,0; AND them $\rightarrow$ 0,1,1,0.) (1 mark per correct pair of rows = up to 2; allow follow-through.) F is 1 only when the inputs **differ**, so it is equivalent to an **XOR** gate (2 marks: 1 for the column completed correctly, 1 for identifying XOR).

Q10 — [4] (AO3) — 1.4.2 (a) First **pop** returns **5**; second **pop** returns **9** (1 + 1). Trace: push 8 $\rightarrow$ [8]; push 3 $\rightarrow$ [8,3]; push 5 $\rightarrow$ [8,3,5]; **pop** $\rightarrow$ **5** $\rightarrow$ [8,3]; push 9 $\rightarrow$ [8,3,9]; **pop** $\rightarrow$ **9** $\rightarrow$ [8,3]. (b) Stack contents bottom $\rightarrow$ top: **8, 3** (1). A stack is **LIFO (Last In, First Out)** — the **most recently pushed item is the first popped**, as both pops returned the most recent pushes (1).

Q11 — [4] (AO3) — 1.4.2 (a) BST from inserting 60, 40, 80, 30, 50, 70:



(1 for 60 root with 40 left / 80 right; 1 for 30 & 50 under 40 and 70 as left child of 80.) (b) In-order (left, root, right): **30, 40, 50, 60, 70, 80** (1). This is **ascending sorted order**, demonstrating the BST ordering property (left subtree < node < right subtree) (1).

Q12 — [6] (AO3) — synoptic 1.1 + 1.4 + 1.3 Mark holistically; up to 6 from a reasoned evaluation covering **all three** decisions: - **Processor:** recommend **RISC** (1) — fixed-length, single-cycle instructions give **low power consumption and predictable, fast real-time response** suited to a battery embedded device; (reject CISC unless its fewer-instructions benefit is justified against the low-power need) (1 for justification). - **Data structure:** recommend a **(circular) queue** (1) to buffer the **most recent readings in time order (FIFO)** with fixed memory and O(1) enqueue/dequeue; (accept a justified stack/array only with sound reasoning) (1 for justification). - **Compression:** recommend **lossless** (1) because **medical readings must not lose accuracy/data**, so lossy (which discards data) is unsuitable despite smaller files (1 for justification linking to clinical accuracy). *Indicative full-mark answer weaves RISC (power/real-time) + circular queue (recent readings, FIFO, fixed memory) + lossless (data integrity), each justified against the heart-rate monitor's requirements.*